COMPUTER-AIDED DESIGN (CAD), ENGINEERING

OVERVIEW

Applying leadership and 21st century skills, participants use complex computer graphic skills, tools, and processes to develop three (3)-dimensional representations of engineering subjects such as a machine part, tool, device, or manufactured product.

ELIGIBILITY

Two (2) individuals per state may participate.

Participants may compete in either CAD, Architecture or CAD, Engineering, but not both events.

TIME LIMITS

A. Thirty (30) minutes are allowed for set-up time.
B. Three (3) hours are allowed for participants to develop drawing(s).
C. Two (2) minutes will be allowed for an interview.

ATTIRE

TSA competition attire is required for this event.

PROCEDURE

A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules.
B. Participants review the Use of Generative Artificial Intelligence (GenAI) found in the General Rules.
C. Participants bring their own computer systems (see Regulation A) to the event area at the time and place stated in the conference program.
D. Each participant, with one (1) assistant (an instructor, fellow student, or adult chaperone), is allowed to set up and test the equipment. At the end of the thirty (30) minute set-up period, assistants are required to leave the area.
E. Participants are given a design problem to solve in a three (3) hour work session.
F. Participants work independently, without assistance from judges, teachers, or fellow participants.
G. Participants are advised to save their work on their hard drives every fifteen (15) minutes.
H. At the end of the session, participants save their work on their hard drives and save the solution in PDF format for an online submission.
I. Judges circulate to evaluate the progress of the participants.
J. At the completion of the three (3) hour time to develop drawings students will upload the solution PDF using the instructions provided by the event coordinator. Solution can be uploaded by the participant early if the full time is not needed.
K. Once the solution is uploaded, the participant can remove all computer equipment without disturbing the remaining participants that are still working.
L. Each participant will sign up for a two (2) minute interview that will take place during the time and location printed in the conference program.
M. Judging and interview will only use the PDF uploaded during the completion of the on-site problem and will be judged on a judge's computer.
N. The top ten (10) finalists are announced during the award ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

A. Participants provide their own computer systems including:
    1. Computer hardware (only one [1] CPU and one [1] monitor); laptops are recommended

2. Software needed for the challenge. Preference is given to CAD software installed directly on the device. Web-based CAD programs are permitted; however, participants must supply their own hotspot for access. National TSA does not provide Internet access and does not guarantee reliable connectivity during the event.
3. Power strip/surge protector
4. Reference materials:
   a. Hard copy recommended
   b. Access to Internet will not be provided. If using student provided hotspot, National TSA does not guarantee reliable connectivity during the event.

B. Internet access is permitted and required for the on-site PDF submission, but it will not be provided by TSA. TSA cannot guarantee the availability of cellular and/or Wi-Fi signals in the competition room.

C. Participants save the solution for the on-site problem as a PDF and submit their solution using the submission procedures provided by the event coordinator.

D. Participants may leave the room only with permission from the event coordinator.

E. A table, chair, sketching paper, and electricity is supplied for each participant.

F. Participants are required to provide their own pencils.

G. Participants are not permitted to share solutions to problems, reference materials, hardware, or software.

H. Participants identify their work using only their student identification number.

I. File name for your solution PDF must be only your student identification number.

J. Once the participant submits the PDF, the participant will sign up for a two (2) minute interview time. Interviews will take place at the time and location printed in the conference program.

K. Evaluation will use the solution PDF submitted on-site using the judge's laptop.

L. Depending on the on-site challenge, multiple PDFs may be required. Unless told otherwise, the solution must be one (1) PDF. If multiple PDFs are required, the student identification number is used for each PDF and add a letter to the end of each PDF with the first PDF having an "a."

M. Participant will not need to bring a computer or any equipment to the interview.

EVALUATION

A. Design solution evaluated on screen using the submitted PDF according to the criteria on the official rating form.

B. Interview

Refer to the official rating form for more information.

GENERATIVE ARTIFICIAL INTELLIGENCE

Use of Generative Artificial Intelligence (GenAI) is NOT permitted for this event.

STEM INTEGRATION

This event aligns with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Engineer
- Automobile designer
- CAD professional
- Machine designer

Participant/Team ID# _____

CAD, ENGINEERING

2027 & 2028 OFFICIAL RATING FORM

HIGH SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ Computer hardware is present
☐ CAD Software
☐ ENTRY NOT EVALUATED

SOLUTION TO PROBLEM (70 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Accuracy (X2)	Solution demonstrates a poor understanding of the geometry; All solution elements are accurate and precise, effectively modeling the problem assigned.	Solution demonstrates an some understanding of the geometry; All solution elements are accurate and precise, effectively modeling the problem assigned.	Solution demonstrates an exemplary understanding of the geometry; All solution elements are accurate and precise, effectively modeling the problem assigned.	
Completeness (X1)	Solution models little of the problem assigned.	Solution models most of the problem assigned.	Solution models all portions of the problem assigned.	
CAD Tools & Functions (X2)	Solution demonstrates a poor degree of competency with CAD tools and features.	Solution demonstrates a some degree of competency with CAD tools and features.	Solution demonstrates a high degree of competency with CAD tools and features.	
CAD Features (X1)	Solution demonstrates a poor understanding and application of various special features of CAD.	Solution demonstrates a moderate understanding and application of various special features of CAD.	Solution demonstrates a complete understanding and application of various special features of CAD.	
Model Efficiency (X1)	Solution reflects an poor usage of features, with some sketches, features and tools being unnecessary.	Solution reflects an average usage of features, with some sketches, features and tools being unnecessary.	Solution reflects an economy of features, using only necessary sketches, features, and tools.	
			SOLUTION TO PROBLEM SUBTOTAL (70 points)	

LAYOUT (50 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Title Block (X1)	Title block and other identifying information for the solution is incomplete.	Title block and other identifying information for the solution is mostly complete.	Title block and other identifying information for the solution is complete with all necessary information.	
Views (X1)	Solution utilizes only one view; view limits the effectiveness of evaluation of the solution.	Solution utilizes two to three (2-3) views of the solution; views are basic and allow a standard evaluation of the solution.	Solution utilizes four (4) or more views of the solution; views are exemplary and allow for thorough evaluation of the solution.	
Quality of Solution (X2)	Solution includes minimal visual evidence that highlights the best aspects of the solution to the given problem.	Solution includes basic visual evidence that highlights the best aspects of the solution to the given problem.	Solution is exemplary and provides excellent evidence from multiple angles (that include among other aspects – equal scale and minimizes visual clutter) of the solution to the given problem.	
Scale (X1)	Scale selected for the drawings is incorrect and not properly noted.	Scale selected is generally correct and properly noted for most drawings.	Scale selected for all aspects of the drawings is correct and properly noted.	
			LAYOUT SUBTOTAL (50 points)	

ENGINEERING APPLICATION (20 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Application of Practices (X1)	Many of the engineering practices selected and used are incorrectly applied.	Most of the engineering practices selected and used are correctly applied.	All of the engineering practices selected and used are correctly and appropriately applied.	
Appropriate Use of Conventions (X1)	There is little or no evidence of an effective application of engineering conventions in the completed design and drawings.	There is some evidence of an effective application of engineering conventions in the completed design and drawings.	There is clear evidence of an effective and knowledgeable application of engineering conventions in the completed design and drawings.	
			ENGINEERING APPLICATION SUBTOTAL (20 points)	

INTERVIEW (30 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
CAD Functions (X1)	There is little evidence of an understanding and application of CAD functions.	There is evidence of a general understanding and effective application of CAD functions.	A complete and effective understanding and application of CAD functions is evident.	
CAD Features (X1)	There is little evidence of an understanding and application of CAD special features.	There is a general understanding and application of CAD special features.	There is a complete understanding and application of the various special features of CAD.	
Articulation (X1)	Communication of the solution is unclear, unorganized, and or illogical; leadership and/or 21st century skills are not evident.	Communication of the solution is somewhat logical and clear; leadership and/or 21st century skills are somewhat evident.	Communication of the solution is clear, concise, and logical; leadership and/or 21st century skills are clearly evident.	
			INTERVIEW SUBTOTAL (30 points)	

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. **TOTAL** (170 points)	

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

COMPUTER-AIDED DESIGN (CAD), ENGINEERING EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

A. Event coordinator
B. Judges, two (2) or more
C. Assistants, one (1)

MATERIALS

A. Coordinator's packet, containing:
    1. Event guidelines, one (1) copy for the coordinator and for each judge
    2. TSA Event Coordinator Report
B. Tables and chairs for competitors and judges
C. One (1) ream of 8½" x 11" white copier paper
D. Stopwatch
E. Statement of problem as a hard-copy sketch for each participant, pages as needed
F. Statement of problem as a hard-copy sketch for each judge, pages as needed
G. One (1) laptop to judge the submissions. Laptop must have Wi-Fi and be capable of opening and viewing a PDF from the online submission
H. Interview questions

RESPONSIBILITIES

AT THE CONFERENCE

A. Attend the mandatory coordinator's meeting at the designated time and location.
B. Report to the CRC room and check the contents of the coordinator's packet.
C. Review the event guidelines and check to see that enough personnel have been scheduled.
D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
E. At least one (1) hour before the event is to begin, meet with judges to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
F. Ensure the judges have access to the online judging system.

ON-SITE CHALLENGE

A. As participants arrive, check the coordinator's report and assign participants to work stations.
B. All participants and judges should be in the room at this time.
C. Late entries are considered on a case-by-case basis and only when the delay is caused by events beyond participant control.
D. In order to compete, participants must be on the entry list or must have approval of the CRC.
E. Allow thirty (30) minutes for participants and their assistants (no more than one [1] per participant) to set up equipment.
F. At the end of the thirty (30)-minute set-up time, non-participants are required to leave the event area.
G. Review with the participants the time limits, procedures, regulations, and protocol of the event.
H. Remind participants to save their work at regular time intervals.
I. Explain the on-line submission procedure. Distribute copies of the CAD problem. Answer any appropriate questions concerning the problem.
J. Begin the event and announce the ending time.
K. During the event, the judges and assistants monitor and evaluate participant progress and work.
L. Participants must submit the solution PDF using the correct procedure. Participants may leave the room once finished, but must first sign up for an interview time that will take place during the time and location printed in the conference program.
M. Announce the time remaining to work at one (1) hour, thirty (30) minutes, fifteen (15) minutes, and five (5) minutes before time is called.
N. When time is called, participants must have stopped, saved their work, and correctly submitted the PDF.
O. Conduct the interviews as the submissions are reviewed. Interviews should be a maximum of two (2) minutes in length.

P. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:

   1. To deduct twenty percent (20%) of the total possible points in this round or
   2. To disqualify the entry

   The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

Q. If necessary, manage security and the removal of materials from the area.